

# **Customer Retention & Revenue Analytics for an E-commerce Business**

## **1. Dataset Overview**

Objective: analyze customer purchasing behavior, revenue generation, product performance, customer satisfaction, and operational efficiency using transaction-level data.

## **2. Data Inspection**

Reviewed dataset dimensions, columns, data types, descriptive statistics, and sample records to understand data quality.

## **3. Data Cleaning**

Handled missing values, removed duplicates, converted dates, validated values, and detected outliers using IQR and Z-score techniques.

## **4. Feature Engineering**

Created Month, Year, Day, Age Group, Revenue Category, and Delivery Type features for deeper analysis.

## **5. Exploratory Data Analysis**

Examined sales trends, revenue by category, age distribution, ratings, payment methods, delivery time, return behavior, and correlation heatmaps.

## **6. Revenue Analytics**

Computed total revenue, average order value, revenue by category, payment method, and customer segment.

## **7. Customer Behaviour Analysis**

Customer ID is unavailable, so proxy indicators such as returns, ratings, and delivery performance were used instead of true retention metrics.

## **8. Correlation & Statistics**

Applied descriptive statistics, skewness, kurtosis, correlation analysis, and outlier detection.

## **9. Business Insights**

Summarized major revenue drivers, customer segments, operational issues, and product performance.

## **10. Recommendations & Conclusion**

Improve delivery speed, reduce returns, prioritize profitable categories, personalize marketing, and use insights for data-driven decision making.